Unit 2: Polynomials and Rational Functions

Precalculus

Student

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2.1 Quadratic Functions

Objectives

- Classify polynomial functions by degree.
- Identify the key features of a parabola (vertex, axis of symmetry, intercepts).
- Convert between vertex form and standard form by completing the square.
- Solve quadratic-type equations using substitution.

Polynomial Functions

A polynomial function is classified by its **degree** — the largest exponent on the variable.

Equation	Degree	Type
$f(x) = c$	________	________
$f(x) = mx + b$	________	________
$f(x) = ax^2 + bx + c$	________	________
$f(x) = ax^3 + bx^2 + cx + d$	________	________
$f(x) = ax^4 + bx^3 + cx^2 + dx + e$	________	________

- The **leading coefficient** is the coefficient on the highest-degree term. It controls end behavior and (for quadratics) whether the parabola opens up or down.
- The **constant term** is the term with no variable. It is always the y-intercept: $f(0)$.

The graph of every quadratic function is a **parabola**. It has five features worth naming.

Anatomy of a Parabola

- **Vertex** — the highest or lowest point on the parabola.
- **Axis of symmetry** — the vertical line through the vertex; the parabola is a mirror image on either side.
- ***x*-intercepts** — the points where the parabola crosses the x-axis (may have 0, 1, or 2).
- ***y*-intercept** — the point where the parabola crosses the y-axis.

- If $a > 0$, the parabola opens ________ and the vertex is a ________.
- If $a < 0$, the parabola opens ________ and the vertex is a ________.

Forms of a Quadratic Function

Feature	Vertex Form $a(x-h)^2 + k$	Standard Form $ax^2 + bx + c$
Vertex	______________	______________
Axis of sym.	______________	______________
y-intercept	______________	________

To convert from standard form to vertex form, ________ ________.

Note

Standard Form → Vertex Form (Completing the Square):

1. Factor out a from the x-terms: $a\left(x^2 + \frac{b}{a}x\right) + c$.

2. Take half the inner coefficient, square it: $\left(\frac{b}{2a}\right)^2$.
3. Add and subtract that value inside the parentheses.
4. Factor the perfect-square trinomial and simplify.

Example 1

Given $f(x) = 2x^2 + 8x + 7$, write the function in vertex form, identify the vertex, and sketch the graph.

Example 2

Write the standard form of the equation of the parabola whose vertex is (1, 2) and that passes through the point (3, −6).

Example 3

Sophia has a food truck and sells tacos. Her daily costs are approximated by $C(x) = x^2 - 40x + 610$, where $C(x)$ is the cost in dollars to sell x tacos.

(a) Find the number of tacos she must sell to minimize her cost.

(b) What is the minimum cost?

Quadratic-Type Equations

A polynomial that *isn't* degree 2 can still factor like a quadratic. Look for three clues:
1. Exactly **three terms** (a trinomial).
2. The exponent on the first term is ________ the exponent on the second.
3. The third term is a ________.

Strategy: Substitute u for the middle-term variable expression (e.g., $u = x^5$). The equation becomes

$au^2 + bu + c = 0$ — a familiar quadratic. Solve for u, then back-substitute to solve for x.

Example 4

Solve $x^6 + 7x^3 = 8$.

Homework: Quadratic Functions Worksheet

2.2 Graphs of Polynomial Functions

Objectives

- Identify the features that make a graph polynomial (continuous, smooth).
- Predict end behavior from the leading term.
- Use multiplicity to determine whether the graph crosses or is tangent to the x-axis.
- Sketch polynomials from factored form.

What Makes a Graph Polynomial

- The graph of every polynomial function is ________ — no breaks, no holes, no jumps.
- The graph has only ________ ________ — no sharp corners or cusps.
- ________ properties must hold for the graph to be polynomial. Either one alone is ________ ________.

Zeros and Turning Points

For a polynomial function f of degree n:
- f has at most ________ real zeros (i.e., x-intercepts).
- The graph of f has at most ________ turning points.

A **zero** of f is a number x for which ________. Equivalently, it is an x-intercept of the graph.

End Behavior and the Leading Coefficient Test

End behavior describes what happens as x becomes large in the ________ ________ direction. It is determined by the function's ________ and the ________ of its leading coefficient.

Case	Degree	Leading Coef.	End Behavior
1	even	positive	both ends ↑
2	even	negative	both ends ↓
3	odd	positive	left ↓, right ↑
4	odd	negative	left ↑, right ↓

Example 1

Which of the following could be the graph of a polynomial whose leading term is $-3x^4$?

Example 2

Describe the end behavior of each polynomial.

a) $f(x) = -x^3 + 4x$

b) $f(x) = x^4 - 5x^2 + 4$

Repeated Zeros and Multiplicity

A factor of $(x-a)^k$ with $k > 1$ means $x = a$ is a **zero of multiplicity** k. The multiplicity controls how the graph meets the x-axis at that zero:

- If k is ________, the graph ________ the x-axis at $x = a$.
- If k is ________, the graph is ________ ________ the x-axis at $x = a$ (touches and bounces back).
- The higher the multiplicity, the ________ the graph passes through (or past) the x-axis.

Example 3

For $f(x) = -(x-1)(x+2)(x-3)$:

(a) Identify the degree. (b) Identify the x-intercepts. (c) Cross or tangent at each? (d) Sketch.

Example 4

For $f(x) = (x+2)^2(x-3)^3$:

(a) Identify the degree. (b) Identify the x-intercepts. (c) Cross or tangent at each? (d) Sketch.

Example 5

How many turning points can each polynomial have at most?

a) $f(x) = -(x+1)^2(x-3)$

b) $f(x) = (x+1)^2(x-3)^3$

Example 6

Write an equation for a polynomial graph with x-intercepts at $x = -3, -1, 1, 2$ whose left and right arms both point **down**. Assume the leading coefficient is 1 or -1.

Homework: Graphs of Polynomial Functions Worksheet

2.3 Zeros of Polynomial Functions

Objectives

- Translate fluently among zero, root, solution, factor, and x-intercept.
- Sketch polynomial graphs by factoring and using multiplicities.
- Build a polynomial from a given set of zeros.
- Apply the Intermediate Value Theorem to guarantee zeros.

Five Names, One Idea

If $f(x)$ is a polynomial and a is a real number, the following five statements are equivalent — they all mean the same thing:

- $x = a$ is a ________ of the **function** $f(x)$.
- $x = a$ is a ________ of the polynomial **equation** $f(x) = 0$.
- $(x - a)$ is a ________ of the **polynomial** $f(x)$.
- $(a, 0)$ is an ________ of the **graph** of f.

- Sometimes zeros are called ________.

Example 1

For $f(x) = x^3 - x^2 - 2x$, use the zeros, y-intercept, and end behavior to sketch the graph.

Example 2

For $g(x) = x^3 - 2x^2 - 4x + 8$, factor and sketch.

Example 3

For $f(x) = -2x^4 - x^3 + 3x^2$, factor and sketch.

Example 4

For $f(x) = x^4 - 12x^2 + 27$, find all real zeros and sketch.

Building a Polynomial from Its Zeros

If a polynomial has zeros $r_1, r_2, \ldots, r_n$, then: $f(x) = a\,(x - r_1)(x - r_2)\cdots(x - r_n)$

- A zero listed twice gives a factor with exponent 2 (its multiplicity).

- For a fractional zero like $-\frac{1}{2}$, use the factor $(2x + 1)$ to keep coefficients as integers.
- If $p + \sqrt{q}$ is a zero, then $p - \sqrt{q}$ is also a zero (conjugate radicals).

Example 5

Write an equation of a polynomial with zeros $-\frac{1}{2},\ 3,\ 3$.

Example 6

Write an equation of a polynomial with zeros $3,\ 2 + \sqrt{3},\ 2 - \sqrt{3}$.

Intermediate Value Theorem (IVT)

If f is a polynomial (continuous) and ________ while ________ (or vice versa), then f has **at least one zero** between a and b.

- A continuous function cannot jump over the x-axis.
- IVT proves a zero *exists* but does not tell you *where*.
- If $f(a)$ and $f(b)$ have the same sign, the test is inconclusive.

Example 7

Show that $f(x) = x^4 - 4x + 1$ has at least one zero between $x = 0$ and $x = 1$.

Homework: Zeros of Polynomial Functions Worksheet

2.4 Dividing Polynomials

Objectives

- Divide polynomials using long division.
- Divide polynomials using synthetic division (linear divisors only).
- Apply the Remainder Theorem and the Factor Theorem.
- Use the Rational Zero Test to list possible rational zeros.

Polynomial Long Division

Long division of polynomials works exactly like long division of whole numbers: **divide, multiply, subtract, bring down, repeat**.

The result always has the form:

$$\frac{f(x)}{d(x)} = q(x) + \frac{r(x)}{d(x)}$$

where $q(x)$ is the ________ and $r(x)$ is the ________.

- The degree of the remainder is always ________ than the degree of the divisor.
- Always write polynomials in ________ ________ of degree before dividing.
- Use 0 as a placeholder for any missing powers.

Example 1

Divide $2x^3 + 3x^2 - 5x + 1$ by $x + 2$ using long division.

Example 2

Divide $6x^3 + 10x^2 + x + 8$ by $2x^2 + 1$.

Synthetic Division

Synthetic division is a shortcut for dividing by a **linear** divisor of the form ________.

Steps:

1. Write c on the left and the coefficients of the dividend across the top.
2. Bring down the first coefficient.
3. Multiply by c, write the product under the next coefficient, and add.
4. Repeat until you reach the last column — that entry is the ________.

The bottom row (excluding the remainder) gives the coefficients of the quotient, whose degree is **one less** than the dividend's.

Example 3

Use synthetic division to divide $x^3 - 7x + 6$ by $x - 2$.

Example 4

Use synthetic division to divide $3x^4 - 5x^3 + 4x - 6$ by $x - 1$.

The Remainder Theorem

If a polynomial $f(x)$ is divided by $(x - c)$, then the remainder equals ________.

In other words: evaluating $f(c)$ and dividing by $(x - c)$ give the same number — the remainder.

The Factor Theorem

$(x - c)$ is a factor of $f(x)$ if and only if ________.

This is just the Remainder Theorem with the remainder equal to zero. It connects factoring and root-finding.

Example 5

Use the Remainder Theorem to evaluate $f(3)$ for $f(x) = 2x^3 - 5x^2 + x - 3$.

The Rational Zero Test

If a polynomial $f(x) = a_n x^n + \cdots + a_1 x + a_0$ has integer coefficients, then every rational zero has the form:

$$\frac{p}{q}$$

where p is a factor of the ________ ________ a_0 and q is a factor of the ________ ________ a_n.

Strategy: List all $\frac{p}{q}$ combinations, then test each candidate using synthetic division or direct evaluation.

Example 6

List all possible rational zeros of $f(x) = 2x^3 + 3x^2 - 8x + 3$, then find all actual zeros.

Homework: Dividing Polynomials Worksheet

2.5 Complex Numbers

Objectives

- Define the imaginary unit i and simplify powers of i.
- Write complex numbers in standard form $a + bi$.
- Add, subtract, multiply, and divide complex numbers.
- Solve quadratic equations with complex solutions.

The Imaginary Unit

The imaginary unit i is defined by:

$$i = \underline{\hspace{2cm}}, \qquad i^2 = \underline{\hspace{2cm}}$$

For any positive real number a:

$$\sqrt{-a} = \underline{\hspace{2cm}}$$

Standard form of a complex number: ________, where a is the ________ part and b is the ________ part.

Example 1

Write each number in standard form.

a) $\sqrt{-9}$ b) $\sqrt{-48}$ c) $3 + \sqrt{-25}$

Operations on Complex Numbers

Let $a + bi$ and $c + di$ be complex numbers.

- **Addition:** $(a + bi) + (c + di) =$ ________
- **Subtraction:** $(a + bi) - (c + di) =$ ________
- **Multiplication:** Use FOIL, then replace i^2 with -1.

Example 2

Simplify. Write answers in standard form.

a) $(3+2i)+(5-4i)$

b) $(7-i)-(3+2i)$

Example 3

Multiply. Write answers in standard form.

a) $(2+3i)(4-i)$

b) $(3+2i)^2$

Complex Conjugates

The **complex conjugate** of $a+bi$ is ________.

Key property: $(a+bi)(a-bi)=a^2+b^2$ — always a ________ ________.

To **divide** complex numbers, multiply the numerator and denominator by the ________ ________ of the denominator.

Example 4

Divide. Write in standard form: $\dfrac{2+3i}{4-i}$

Powers of i

The powers of i cycle with period ________:

$$i^1=i,\qquad i^2=-1,\qquad i^3=-i,\qquad i^4=1,\qquad i^5=i,\ \ldots$$

To simplify i^n: divide n by 4 and use the ________.

Example 5

Simplify: a) i^{23} b) i^{42} c) i^{100}

Example 6

Solve $x^2 + 4x + 7 = 0$.

Homework: Complex Numbers Circuit Worksheet

2.6 Fundamental Theorem of Algebra

Objectives

- Use the Fundamental Theorem of Algebra to count zeros.
- Factor any polynomial completely over the complex numbers.
- Apply the Conjugate-Zero Theorems to recover missing zeros.
- Build polynomials with given zeros and an additional condition.

The Fundamental Theorem of Algebra (FTA)

Every polynomial equation with degree greater than zero has ________ ________ ________ in the set of complex numbers (counting multiplicity).

The polynomial therefore factors completely as a product of ________ ________ over $\mathbb{C}$:

$$f(x) = a\,(x - r_1)(x - r_2)\,\cdots\,(x - r_n)$$

- A degree-n polynomial has n zeros in $\mathbb{C}$.
- A repeated zero of multiplicity k counts as k of those zeros.
- Over the *real* numbers, a polynomial can have fewer than n zeros. Over $\mathbb{C}$, the count is exact.

Example 1

Factor $P(x) = x^3 + x^2 - 4x + 6$ completely over $\mathbb{C}$.

Example 2

Find all zeros of $f(x) = x^5 + 6x^3 + 2x^2 - 27x + 18$.

Conjugate-Zero Theorems

When a polynomial has **real coefficients**, non-real and irrational zeros come paired with their conjugates:

- **Complex Conjugate Zeros:** If $a + bi$ is a zero, then ________ is also a zero.
- **Irrational Conjugates:** If $a + \sqrt{b}$ is a zero, then ________ is also a zero.

Non-real zeros always pair up — a degree-n polynomial with real coefficients has an **even** number of non-real zeros.

Multiplying a conjugate pair gives a quadratic with real coefficients:

$$(x - (a + bi))(x - (a - bi)) = (x - a)^2 + b^2$$

Example 3

Write a cubic polynomial g with real coefficients that has zeros 2 and $\sqrt{3}$, and where $g(4) = 13$.

Example 4

Find a cubic polynomial f with real coefficients that has 2 and $1 - i$ as zeros, and $f(1) = 3$.

Example 5

Given that $1 + 3i$ is a zero of $f(x) = x^4 - 6x^3 + 47x^2 - 98x + 290$, find all zeros.

Homework: Fundamental Theorem of Algebra Worksheet

2.7 Rational Functions and Asymptotes

Objectives

- Define rational functions and their domains.
- Identify vertical and horizontal asymptotes.
- Distinguish between holes and vertical asymptotes.
- Compare degrees to determine horizontal asymptotes.

Rational Functions

A **rational function** is a quotient of two polynomials:

$$f(x) = \frac{g(x)}{h(x)}, \qquad h(x) \neq 0$$

The **domain** excludes every x-value that makes the denominator equal to ________.

The parent rational function is $f(x) = \dfrac{1}{x}$, a ________ with domain $x \neq 0$ and range $y \neq 0$.

Vertical and Horizontal Asymptotes

- If $f(x) \to \pm\infty$ as $x \to a$, then $x = a$ is a ________ ________.
- If $f(x) \to a$ as $x \to \pm\infty$, then $y = a$ is a ________ ________.

Asymptotes and Holes — The Rules

Factor first, cancel common factors to expose holes, then read off the features.

Feature	Condition	How to Find
Vertical Asymptote	Values that make ________ the denominator zero	Set remaining denom. = 0.
Horiz. Asymptote	deg(num) ________ deg(denom)	$y = 0$
	deg(num) ________ deg(denom)	$y = \dfrac{\text{leading coef. of num}}{\text{leading coef. of denom}}$
Hole	Common factors in ________ num. and denom.	Set the common factor = 0.
Slant Asymptote	deg(num) ________ deg(denom) (by exactly 1)	Long/synthetic division; quotient is the slant line.

Example 1

Find the domain of $f(x) = \dfrac{1}{x-1}$ and describe the behavior near any excluded x-values.

Example 2

Find the asymptotes and/or holes.

a) $f(x) = \dfrac{1}{2x}$

b) $f(x) = \dfrac{2x^2}{x^2-1}$

Example 3

Find the asymptotes and/or holes.

a) $f(x) = \dfrac{3x}{2x^2+1}$

b) $y = \dfrac{(x+4)(x-2)}{(x-3)(x+4)}$

Example 4

For $y = \dfrac{x^2 + x - 6}{x^2 - x - 2}$: (a) Identify asymptotes and/or holes. (b) Write the domain. (c) Find the intercepts.

Example 5

For $f(x) = \dfrac{3x^3 + 7x^2 + 2}{-2x^3 + 16}$: (a) Identify asymptotes and/or holes. (b) Write the domain.

Example 6

For $f(x) = \dfrac{x^2 - 16}{x^2 + 8x}$: (a) Asymptotes/holes. (b) Domain. (c) Intercepts.

Homework: Rational Functions Worksheet

2.8 Graphs of Rational Functions

Objectives

- Follow a four-step procedure to sketch any rational function by hand.
- Identify and draw slant (oblique) asymptotes.
- Use test points to determine the shape of each branch.

Guidelines for Graphing a Rational Function

Every rational graph follows the same four-step recipe:

1. **Find & sketch asymptotes.** Factor num. and denom. Cancel common factors (holes). Read VA, HA, or slant asymptote. Draw each as a dashed line.
2. **Find & plot intercepts.** x-int from zeros of the (simplified) numerator. y-int is $f(0)$ (if in the domain).
3. **Plot test points.** At least one point between and one beyond each x-intercept and VA.
4. **Connect smoothly.** Approach but never cross VAs. May cross HAs in the middle.

Note

Don't connect across a vertical asymptote. The graph splits into separate branches — sketch each branch independently.

Example 1

Sketch $g(x) = \dfrac{3}{x-2}$.

Example 2

Sketch $f(x) = \dfrac{x^2}{x^2 - x - 2}$.

Example 3

Sketch $f(x) = \dfrac{x^2 - 9}{x^2 - 2x - 3}$.

Slant (Oblique) Asymptotes

When the degree of the numerator is *exactly one more* than the denominator's, the long-run behavior is a tilted line.

Condition: $\deg(\text{num}) = \deg(\text{denom}) + 1$

How to find: Long-divide the numerator by the denominator. The ________ (ignore the remainder) is the slant asymptote.

- A slant asymptote replaces — does **not coexist with** — a ________ asymptote.
- If the numerator degree exceeds the denominator's by more than 1, there is ________ ________.

Example 4

Sketch $f(x) = \dfrac{x^2 + x}{x - 1}$.

Example 5

Sketch $y = \dfrac{x^3}{x^2 - 4}$.

Homework: Graphs of Rational Functions Worksheet